

HOT WORK

Prevent fire and explosion during welding, cutting and grinding.

PURPOSE AND SCOPE

Review fire, explosion, burn and fume hazards associated with welding, cutting, grinding, brazing and other spark-producing work, and verify all required controls before starting. Hot work is not limited to activities involving an open flame. Grinding, metal cutting and other heat-producing work can start fires through sparks, slag and hot surfaces.

KEY HAZARDS

 FIRE	 EXPLOSION	 HOT SLAG	 FUMES
 GAS / VAPOUR	 CYLINDER DAMAGE	 LOWER LEVEL	 HIDDEN EMBER

CRITICAL VALUES TO VERIFY IN THE FIELD

10.7 m

General combustible-clearance area

30 min+

Minimum post-work fire watch

0

Unprotected openings and uncontrolled sparks



REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1926.352 and NFPA 51B address permits, combustible control, fire watch and post-work monitoring together. Apply any stricter site procedure.

OSHA 1926.352(a) FIRE PREVENTION

Remove combustibles, protect items that cannot be moved and prevent sparks travelling through openings.

OSHA 1926.352(e) FIRE WATCH

Where a fire hazard exists, provide a trained, dedicated fire watch and suitable extinguishing equipment.

OSHA 1926.353 VENTILATION

Control welding fumes at source and provide mechanical ventilation in enclosed or restricted spaces.

NFPA 51B PERMIT AND WATCH

Verify permit conditions, the 10.7 m area and the required post-work fire-watch duration.

MINIMUM EVIDENCE BEFORE START

- | | |
|--|--|
| ☐ Approved hot-work permit | ☐ Gas-test record |
| ☐ Fire-watch assignment | ☐ Lower and adjacent-area check |
| ☐ Equipment inspection record | ☐ Post-work monitoring plan |

SUPERVISOR DISCUSSION POINTS

- Who is today's dedicated fire watch?
- Where are the nearest suitable extinguisher and alarm point?
- When was gas testing completed and what were the results?

CONTROLLED HOT-WORK SETUP



Permit - Screens - Fire watch - Extinguisher - Lower-area control

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. CHOOSE A COLD METHOD

Verify the approved hot-work permit. Complete and record required atmospheric testing.

2. PREPARE THE AREA

Remove combustibles or protect them with suitable covers. Inspect and barricade lower and adjacent areas. Assign a trained and dedicated fire watch.

3. CONTROL THE WORK

Provide appropriate extinguishers and fire blankets. Inspect cables, hoses, regulators and equipment. Provide adequate ventilation and fume control.

4. MONITOR AFTER WORK

Contain sparks, slag and hot materials. Confirm the post-work monitoring period on the permit.

CRITICAL WARNING



Some hot-work fires begin after the task is complete because of hidden embers or heated materials.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|---|--|
| ☐ Permit approved | ☐ Gas test acceptable |
| ☐ Fire watch ready | ☐ Extinguisher ready |
| ☐ Combustibles removed | ☐ Lower area checked |
| ☐ Equipment inspected | ☐ PPE suitable |
| ☐ Ventilation adequate | |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			